

CryptoPulse: Final Project Presentation

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Recap: The Problem

- Predicting crypto prices is challenging due to high volatility.
- **Hypothesis:** Sentiment analysis provides a leading indicator for price movements.

- **Market Data (Quant):** Yahoo Finance (BTC-USD, ETH-USD).
- **Sentiment Data (Qual):**
 - ****News Sentiment:**** NLP analysis of headlines.
 - ****Fear & Greed Index:**** Real-time market emotion metric (0-100).

Enhanced User Interface (Live Demo)



Figure: CryptoPulse Market Intelligence Terminal (v2.0)

Experimental Results

Model	MSE	Improvement
Baseline (Naive)	0.051	-
Unimodal (Price)	0.042	+17.6%
Multimodal (Price + FNG)	0.032	+37.3%

Table: Multimodal Fusion significantly outperforms baseline models.

① Conclusion:

- Successfully fused quantitative (Price) and qualitative (Sentiment) data.
- Achieved **0.032 MSE**, a significant improvement over baseline.
- Delivered a premium, interactive "Market Intelligence Terminal".

② Future Directions:

- Integrate **Transformer/Attention** mechanisms for better long-term forecasting.
- Connect to exchange APIs for **Automated Trading**.
- Incorporate real-time social media firehose (e.g., X/Twitter).

Thank You!

Project Repository:

<https://github.com/nizamkadirteach/cdsproject>